atmosphere. The biosphere includes all living organisms on earth, together with the dead organic matter produced by them. The biosphere is absent at extremes of the North and South poles, the highest mountains and the deepest oceans since existing hostile conditions there do not support life (life is the characteristic feature of the biosphere). Occasionally spores of fungi and bacteria do occur at a great height beyond 8000 metres, but they are meta- bolically inactive, and hence represent only dormant life.

33.

- Of the visible part of the spectrum, only red and blue are effective in photosynthesis.
- Plants grown in blue light are small, red light results in elongation of cells (etiolated plants). Plants grown in ultraviolet and violet light are dwarf.
- Frost results in freezing the soil moisture. The plants are killed due to increased transpiration when their roots are unable to supply moisture. Water in the intercellular spaces of the plant gets frozen into ice. This results in increasing concentration of salts and dehydration of cells.

34.

- An ecosystem is a community of organisms interacting with each other and with their environment such that energy is exchanged and system-level processes, such as the cycling of elements, emerge.
- A biome is a large naturally occurring community of flora and fauna occupying a major habitat. For example, rainforest biome or tundra biome.
- Population is a community of interbreeding organisms (same species), occupying a defined area during a specific time.
- Species are a group of living organisms consisting of similar individuals capable of exchanging genes or of interbreeding.

35. An ecotone is a zone of junction or a transition area between two biomes (diverse ecosystems). Ecotone is the zone where two communities meet and integrate. For example, the mangrove forests represent an ecotone between marine and terrestrial ecosystem. Other examples are grassland (between forest and